Capstone Project: Spotify

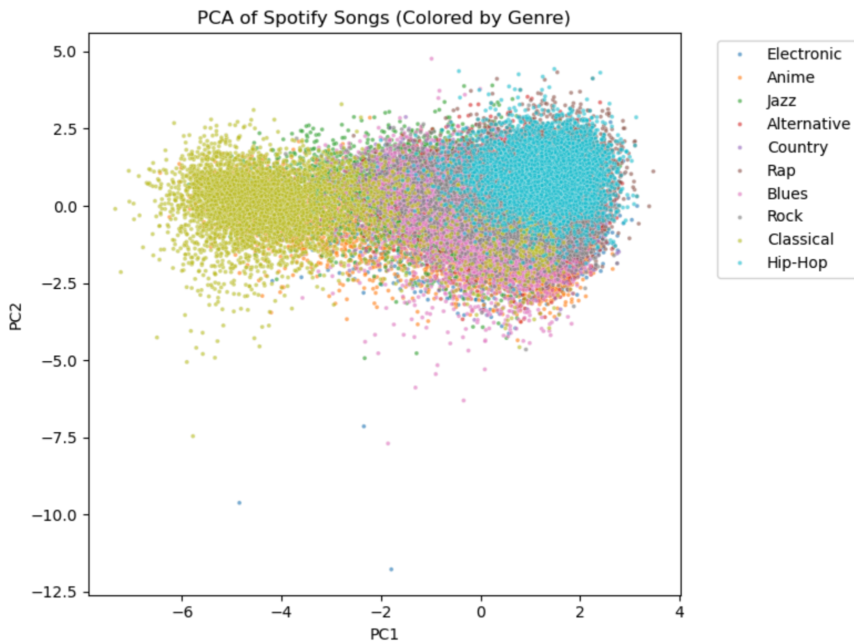
The Process

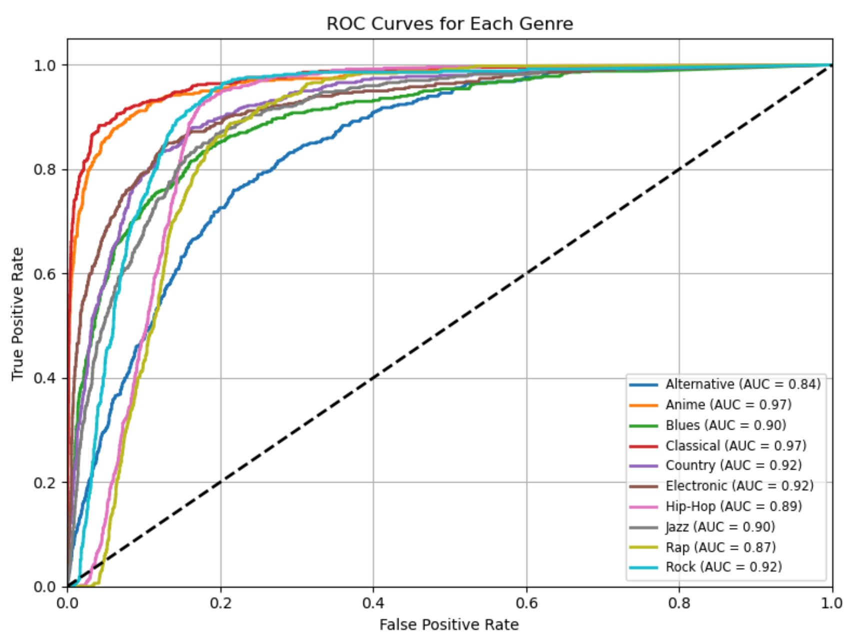
The first step was to load and preprocess the data. I removed the columns that I believed would not hold the most predictive value and then handled missing values by imputing numerical features with their mean and categorical features with the most frequent value. Numerical columns were scaled with StandardScaler, and categorical columns were one-hot encoded. Pipelines were used to better simplify this process. This preprocessing ensured that the data was properly formatted for the classifier. The dataset was split into training and testing sets by genre to ensure that both training and test sets contained samples from each genre with a minimum of 500 samples per genre.

To reduce the dimensionality of the data and remove correlated features, PCA was applied to the preprocessed data. I extracted the first two components, providing a 2D visualization of the data. This reduced complexity and allowed for better insights into the distribution of genres in the feature space. The PCA visualization can be seen below. A better understanding and examination of what the components actually represent is provided in the extra credit portion below. The model was trained using a Random Forest using One-vs-Rest, which trains a separate classifier for each genre against all the others. The ROC curve was plotted for each genre, and the AUC score was computed for all classes.

Results and Conclusion

The model achieved a final weighted average AUC of 0.9112 (the individual AUC scores are shown below), showing strong classification performance. The ROC curves for each genre varied quite a bit, with some genres like “Classical” being more accurately predicted than others. The ROC curve is shown below, alongside the PCA visualization.





In conclusion, the Random Forest Classifier with PCA for dimensionality reduction performed well for the genre classification task. This makes sense as it is capable of non-linear modelling and naturally works well with mixed types of features. The preprocessing, including scaling and encoding, helped standardize the data. The model's final AUC score suggests that the model generalizes well, though classification for some genres could be further optimized.

Extra Credit

I performed PCA to better understand which audio features contributed most to the variance across genres. By examining the PCA component loadings, I found:

- ➢ PC1 is strongly influenced by loudness, energy, and acousticness
- ➢ PC2 is influenced by danceability and tempo.
- ➢ Popularity and danceability are quite correlated. This makes sense, as danceable music is more likely to be replayed, thus making it popular.

This suggests that genres with high energy and loudness (EDM or Metal) are clearly distinguishable from "softer" or acoustic genres (like Jazz or Classical) in PC1 space. A graph is shown below: